When the bucket is kicked flat: Resultatives and the time course of idiom processing

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Introduction: Debates about idiom processing boil down to the problem of timing: at what point does the idiomatic construal become favored and how quickly is that preference reweighted as new cues arrive? While strong "literal first" accounts predict delayed idiom recognition [1-3], many current accounts posit incremental cue-based competition between idiomatic and literal construals, with various factors (e.g., lexical, discourse, syntactic) continuously shifting their relative activation [4-6]. We test this incremental updating view using a novel diagnostic: resultative secondary predicates (e.g., kicked the bucket flat), which are compatible with a literal event structure but incompatible with idiomatic construals [7-9]. In addition, we manipulate determiner type (article vs. demonstrative), which is a grammatical- and discourse-level cue that can modulate the availability of figurative interpretations [10-11].

Methods: A L(exicality)-maze experiment (n=83, 24 item sets) was conducted where Meaning (Idiom vs. Literal) and Determiner (Article vs. Demonstrative) were manipulated as independent factors in a 2x2 factorial design (1). All experimental items shared the same context and resultative predicate within item sets. Idioms were non-compositional expressions whose figurative meanings are not transparently composed from their parts (e.g., *kick ≠ die*; *bucket ≠ suddenly*) and were verified as attested in large corpora (COCA, Google Books). Experimental items were separated by 72 structurally diverse filler sentences. Analyses targeted two time windows: the object noun (where an idiom analysis becomes highly probable) and the resultative (where literal and idiom analyses diverge in compatibility). If cues are integrated incrementally, Determiner type should affect processing at the object noun, and a selective bias on idiomaticity would appear as an interaction effect. In addition, the resultative should trigger rapid updating when it conflicts with an idiom-favored event structure, resulting in a cost at the resultative region.

Results: At the noun region, linear mixed-effects models revealed significant main effects of Meaning ($p<.001$; Idiom < Literal) and Determiner ($p<.001$; Article < Demonstrative), consistent with early cue-based weighting during incremental interpretation, but no reliable interaction ($p=.250$). Consistent with these effects, planned pairwise comparisons showed that within Articles, Idiom sentences were read faster than Literal sentences ($p=.001$), and within Idioms, Articles were faster than Demonstratives ($p<.001$). At the resultative region, there was a main effect of Meaning ($p=.013$; Literal < Idiom), consistent with an increased processing cost when the resultative forces updating away from the idiomatic construal.

Discussion: The timing of these preliminary results supports a cue-based updating account. The noun-region advantage for Idioms indicates that idiomatic construals can be strongly supported early in processing rather than only after a delayed literal stage. The concurrent determiner effect at the noun region also shows that discourse and grammatical cues impact processing in the same early time window. The reversal at the resultative region, where Idioms were read slower than Literal sentences, suggests the parser has rapid sensitivity to event-structural constraints that are incompatible with a canonical idiomatic construal, yielding immediate reweighting toward a literal analysis. Overall, the pattern is better captured by a parallel, constraint-based approach in which idiomatic and literal interpretations are co-activated and continuously reweighted as cues are encountered during incremental comprehension.

(1) a. *Literal Article*:
While he struggled with the overheating equipment, Carl **bumped a** gasket <u>loose</u> inside the old generator.

b. *Literal Demonstrative*:
While he struggled with the overheating equipment, Carl **bumped this** gasket <u>loose</u> inside the old generator.

c. *Idiom Article*:
While he struggled with the overheating equipment, Carl **blew a** gasket <u>loose</u> inside the old generator.

d. *Idiom Demonstrative*:
While he struggled with the overheating equipment, Carl **blew this** gasket <u>loose</u> inside the old generator.

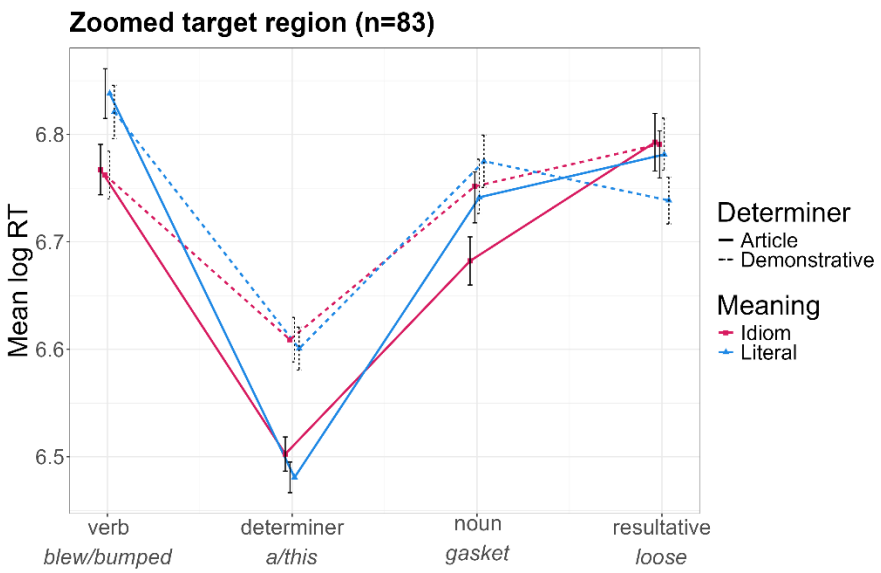

Figure 1. The mean log RTs at the possible idiom and resultative regions.

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